

Chapter 7: Thermal Properties of Matter

Key Concepts

Temperature — Degree of hotness or coldness of a body measured on a definite scale. SI unit: Kelvin (K).

Heat — Form of energy that flows from a hotter body to a colder body due to temperature difference. SI unit: Joule (J).

Thermometer — Device used to measure temperature.

- **Liquid-in-glass** — Uses expansion of mercury/alcohol. Range: -10°C to 110°C .

- **Bimetallic** — Uses differential expansion of two bonded metals. Used in thermostats.

- **Electric/Resistance** — Uses change in electrical resistance with temperature (platinum resistance thermometer). Range: -200°C to 1200°C .

Temperature Scales — 0°C (freezing) to 100°C (boiling) of water.

- **Fahrenheit ($^{\circ}\text{F}$)**: 32°F (freezing) to 212°F (boiling) of water.

- **Kelvin (K)**: 0 K = absolute zero (-273.15°C). No negative temperatures.

Conversion Formulas — $^{\circ}\text{C} = (\text{K} - 273)/100$

- $\text{K} = \text{C} + 273$

- $\text{F} = (9/5)\text{C} + 32$

Specific Heat Capacity (c) — Heat required to raise temperature of 1 kg of a substance by 1 K (or 1°C). $Q = mc\Delta\theta$. Unit: $\text{J kg}^{-1}\text{K}^{-1}$.

Thermal Expansion — $\Delta L = L\alpha\Delta\theta$ (α = linear expansivity)

- **Areal expansion**: $\Delta A = A\beta\Delta\theta$ ($\beta = 2\alpha$; for isotropic solids)

- **Volume expansion**: $\Delta V = V\gamma\Delta\theta$ ($\gamma = 3\alpha$; for isotropic solids)

Latent Heat — Heat absorbed or released during a phase change without temperature change.

- **Latent heat of fusion (L_f)**: Heat to change 1 kg solid to liquid at melting point. Ice: $3.336 \times 10^5\text{ J kg}^{-1}$.

- **Latent heat of vaporization (L_v)**: Heat to change 1 kg liquid to gas at boiling point. Water: $2.26 \times 10^6\text{ J kg}^{-1}$.

Evaporation — Vaporization at the surface of a liquid below its boiling point.

Boiling — Vaporization throughout the bulk of a liquid at a fixed temperature (boiling point).

Heat Transfer — **Conduction**: Transfer through direct molecular collisions in solids. Rate depends on thermal conductivity k .

- **Convection**: Transfer by bulk motion of fluids (liquids/gases).

- **Radiation**: Transfer by electromagnetic waves. No medium required.

Thermal Conductivity (k) — Measure of a material's ability to conduct heat. $Q/t = kA\Delta\theta/l$. Unit: $\text{W m}^{-1}\text{K}^{-1}$.

Greenhouse Effect — Trapping of infrared radiation by atmospheric gases (CO_2 , CH_4 , H_2O vapour), warming the Earth's surface.

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MCQ's

1. The SI unit of temperature is:

A. Celsius B. Fahrenheit C. Kelvin D. Joule

Answer: C

2. The temperature at which all molecular motion ceases is called:

A. 0°C B. -100°C C. Absolute zero D. Freezing point

Answer: C

3. A bimetallic thermometer works on the principle of:

A. Equal expansion of metals B. Differential expansion of metals C. Change in resistance D. Liquid expansion

Answer: B

4. Normal human body temperature is 37°C . What is it in Kelvin?

A. 236 K B. 300 K C. 310 K D. 373 K

Answer: C ($37 + 273 = 310\text{ K}$)

5. The specific heat capacity of water is approximately:

A. $4200\text{ J kg}^{-1}\text{K}^{-1}$ B. $2100\text{ J kg}^{-1}\text{K}^{-1}$ C. $900\text{ J kg}^{-1}\text{K}^{-1}$ D. $138\text{ J kg}^{-1}\text{K}^{-1}$

Answer: A

- **Volume expansion:** Increase in volume: $\Delta V = \gamma V \Delta T$, where $\gamma = \frac{1}{\alpha \beta}$ for isotropic solids.

Q5. Define latent heat of fusion and vaporization.

Ans: Latent heat of fusion: Heat required to change 1 kg of a solid into liquid at its melting point without change in temperature. For ice: $3.336 \times 10^5 \text{ J/kg}$.

Latent heat of vaporization: Heat required to change 1 kg of a liquid into vapor at its boiling point without change in temperature. For water: $2.26 \times 10^6 \text{ J/kg}$.

Q6. Differentiate between evaporation and boiling.

Ans: Evaporation: Occurs at the surface of a liquid at any temperature below boiling point. It is a slow process, causes cooling. Boiling: Occurs throughout the bulk of liquid at a fixed temperature (boiling point). It is a rapid process, temperature remains constant during boiling.

Q7. Explain the three modes of heat transfer with examples.

Ans:

- **Conduction:** Heat transfer by direct molecular collision in solids. Example: iron rod getting hot at one end when the other is in fire.
- **Convection:** Heat transfer by bulk motion of fluid particles. Example: hot air rising from a heater.
- **Radiation:** Heat transfer by electromagnetic waves without a medium. Example: heat from the Sun reaching Earth.

Q8. What is the greenhouse effect? Name two greenhouse gases.

Ans: The greenhouse effect is the trapping of infrared (heat) radiation by certain gases in the Earth's atmosphere, which warms the planet. Two main greenhouse gases: Carbon dioxide (CO_2) and Methane (CH_4).

Q9. Why is water used as a coolant in car engines?

Ans: Water has a high specific heat capacity ($4200 \text{ J/kg}^\circ\text{K}$), meaning it can absorb a large amount of heat without a large rise in temperature. This makes it an effective coolant for removing excess heat from car engines.

Q10. What is a bimetallic thermostat? Where is it used?

Ans: A bimetallic thermostat consists of two different metals (e.g., brass and invar) bonded together. When heated, the metals expand by different amounts, causing the strip to bend. This bending makes or breaks an electrical contact, regulating temperature. Used in: electric irons, ovens, room heaters, refrigerators.

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Long Questions

Q1. Describe different types of thermometers. Compare their working principles, ranges, and uses.

Ans: Three common types of thermometers:

1. Liquid-in-Glass Thermometer:

- **Working:** Based on thermal expansion of a liquid (mercury or alcohol) in a glass capillary tube. As temperature rises, liquid expands and rises in the capillary.
- **Range:** -10°C to 110°C (mercury); -115°C to 78°C (alcohol).
- **Uses:** Clinical (body temperature), laboratory experiments.

2. Bimetallic Thermometer:

- **Working:** Two different metals with different expansivities (e.g., brass and invar) are bonded together. Temperature change causes differential expansion, bending the strip, which moves a pointer.
- **Range:** -50°C to 500°C .
- **Uses:** Thermostats in irons, ovens, refrigerators; industrial temperature control.

3. Electric/Resistance Thermometer (Platinum Resistance):

- **Working:** Based on the principle that electrical resistance of a pure metal (platinum) increases uniformly with temperature. $R_T = R_0(1 + \alpha \Delta T)$
- **Range:** -200°C to 1200°C .
- **Uses:** Accurate industrial measurements, scientific research, very high and very low temperature measurement.

Q2. Derive the relation between different temperature scales. Convert 40°C , 98.6°F , and 300 K to the other two scales.

Ans: Relation between Celsius (C), Fahrenheit (F), and Kelvin (K):

$$C/100 = (F - 32)/180 = (K - 273)/100$$

(a) 40°C to Fahrenheit & Kelvin:

$$F = (9/5)C + 32 = (9/5)(40) + 32 = 72 + 32 = 104^\circ\text{F}$$

$$K = C + 273 = 40 + 273 = 313 \text{ K}$$

(b) 98.6°F to Celsius & Kelvin:

$$C = (5/9)(F - 32) = (5/9)(98.6 - 32) = (5/9)(66.6) = 37^\circ\text{C}$$

$$K = C + 273 = 37 + 273 = 310 \text{ K}$$

(c) 300 K to Celsius & Fahrenheit:

$$C = K - 273 = 300 - 273 = 27^\circ\text{C}$$

$$F = (9/5)C + 32 = (9/5)(27) + 32 = 48.6 + 32 = 80.6^\circ\text{F}$$

Q3. Derive expressions for linear, areal, and volume thermal expansion. Give the relation between their coefficients.

Ans: Linear Expansion:

When a rod of initial length L_0 is heated by ΔT , its length increases by ΔL

$\Delta L = \alpha L_0 \Delta T$, where α is the coefficient of linear expansion.

Final length $L = L_0(1 + \alpha \Delta T)$.

Areal Expansion:

Consider a square sheet of side L_0 , area $A_0 = L_0^2$.

After heating: $L = L_0(1 + \alpha \Delta T)$

New area $A = L^2 = L_0^2(1 + \alpha \Delta T)^2 \approx L_0^2(1 + 2\alpha \Delta T)$ (neglecting $\alpha^2 \Delta T^2$)

So $A = A_0(1 + 2\alpha \Delta T)$

Thus $\Delta A = 2\alpha A_0 \Delta T$, where $\beta = 2\alpha$ = coefficient of areal expansion.

Volume Expansion:

Consider a cube of side L_0 , volume $V_0 = L_0^3$.

After heating: $L = L_0(1 + \alpha \Delta T)$

New volume $V = L^3 = L_0^3(1 + \alpha \Delta T)^3 \approx L_0^3(1 + 3\alpha \Delta T)$

So $V = V_0(1 + 3\alpha \Delta T)$

Thus $\Delta V = 3\alpha V_0 \Delta T$, where $\gamma = 3\alpha$ = coefficient of volume expansion.

Relation: $\beta = 2\alpha$, $\gamma = 3\alpha$ (for isotropic solids).

Q4. Explain the concept of latent heat. Describe an experiment to determine the latent heat of fusion of ice.

Ans: Latent heat is the heat absorbed or released by a substance during a phase change without any change in temperature.

Experiment to find latent heat of fusion of ice (Method of mixtures):

1. Take a calorimeter (copper vessel) of known mass m_c and specific heat capacity c_c .
2. Fill it with water of mass m_w at temperature T_0 .
3. Take dry ice cubes at 0°C of total mass m_i and quickly add them to the calorimeter water.
4. Stir gently until all ice melts and record the final equilibrium temperature T_f .
5. The ice first melts (latent heat) and then the resulting water warms from 0°C to T_f .

Heat gained by ice: $Q = m_i L_f + m_{ic} c_w (T_f - 0)$

Heat lost by calorimeter + water: $Q = m_{cc} c_c (T_0 - T_f) + m_{wc} c_w (T_0 - T_f)$

By the principle of calorimetry (heat lost = heat gained):

$$m_{cc} c_c (T_0 - T_f) + m_{wc} c_w (T_0 - T_f) = m_i L_f + m_{ic} c_w T_f$$

Rearranging:

$$L_f = [m_{cc} c_c (T_0 - T_f) + m_{wc} c_w (T_0 - T_f) - m_{ic} c_w T_f] / m_i$$

Q5. Explain the greenhouse effect. How is it beneficial and what problems does it cause?

Ans: Greenhouse Effect:

The Earth's atmosphere contains greenhouse gases (CO_2 , CH_4 , H_2O vapor, N_2O) that allow short-wavelength solar radiation to pass through but absorb long-wavelength infrared radiation emitted by the Earth's surface. This trapped heat warms the planet, keeping the average temperature around 15°C instead of -18°C .

1. Maintains Earth's temperature suitable for life.
2. Prevents extreme temperature swings between day and night.
3. Enables agriculture and ecosystem stability.

Global warming — Rise in Earth's average temperature due to increased CO₂ from burning fossil fuels and deforestation.

Ecosystem disruption — Loss of biodiversity, coral bleaching.

[illegible]
$$13380 = m_i \times 375600$$

$$m_j = 13380 / 375600 = 0.0356 \text{ kg} = 35.6 \text{ g}$$

Q6. A glass flask has volume 500 cm³ at 20°C. If the coefficient of volume expansion of glass is $2.55 \times 10^{-5} \text{ K}^{-1}$, find its volume at 80°C.

Solution:

Given: $V_1 = 500 \text{ cm}^3$, $T_1 = 20^\circ\text{C}$, $T_2 = 80^\circ\text{C}$, $\gamma = 2.55 \times 10^{-5} \text{ K}^{-1}$

$$V_2 = V_1 + \gamma V_1 \Delta T$$

$$= 500 + (2.55 \times 10^{-5} \times 500 \times 60)$$

$$= 500 + 0.765 \text{ cm}^3$$

$$\text{Volume at } 80^\circ\text{C} = 500.765 \text{ cm}^3$$

Q7. A copper rod of area 0.01 m² and length 0.5 m conducts heat. The temperatures at its ends are 100°C and 30°C. If the thermal conductivity of copper is 400 Wm⁻¹K⁻¹, find the rate of heat flow.

Solution:

Given: $A = 0.01 \text{ m}^2$, $L = 0.5 \text{ m}$, $T_1 = 100^\circ\text{C}$, $T_2 = 30^\circ\text{C}$, $k = 400 \text{ Wm}^{-1}\text{K}^{-1}$

$$\Delta T = 100 - 30 = 70^\circ\text{C}$$

$$\text{Rate of heat flow} = \frac{Q}{t} = \frac{kA\Delta T}{L}$$

$$= \frac{400 \times 0.01 \times 70}{0.5}$$

$$= 400 \times 0.01 \times 140$$

$$= 560 \text{ W}$$

Q8. How much heat energy is required to convert 200 g of water at 100°C into steam at 100°C? ($L_v = 2.26 \times 10^6 \text{ Jkg}^{-1}$)

Solution:

Given: $m = 0.2 \text{ kg}$, $L_v = 2.26 \times 10^6 \text{ Jkg}^{-1}$

$$Q = mL_v = 0.2 \times 2.26 \times 10^6 = 4.52 \times 10^5 \text{ J}$$